

The Weyl-Clock-Cancellation Bridge: Mechanism, Not Leverage

A careful first step connecting discrete proto-phase cancellation to the critical line

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Honesty boundary, stated first because this is RH-adjacent. This note **claims nothing about the Riemann Hypothesis and gives no leverage on the location of the zeros.** The zeros are classical; their values were taken as known and the bouquet was confirmed to cancel there — which it must, since the bouquet *is* the analytically-continued zeta function. The only framework content is the relabeling $\ln n$ = the catalog’s forgetting-costs, and the observation that our discrete proto-phase has its own exact cancellation. Verified by `weyl_clock_cancellation.py`. Claims graded Fact / Reading / Lead.

1. The discrete cancellation seed (exact)

Our proto-phase already contains total destructive interference: the d -th roots of unity sum to zero, to machine precision, for every d . The genesis bit is the simplest case — $1 + (-1) = 0$, the NOT cancellation. Uniform amplitudes, uniform phases $\rightarrow$ exact cancellation. (*Fact.*)

2. The critical-line cancellation is the same mechanism, log-spaced

The resistance-phasor bouquet on the critical line,

$$\eta\left(\frac{1}{2} + it\right) = \sum_{n \geq 1} (-1)^{n-1} n^{-1/2} e^{-it \ln n},$$

is the same idea with **log-spaced phases** $t \cdot \ln n$ and amplitudes $n^{-1/2}$. Computed, $|\text{bouquet}|$ falls to $\sim 10^{-14}$ exactly at the known zeros ($t = 14.135, 21.022, 25.011, 30.425$) and runs $O(1)$ between them; a coarse scan places the nulls at the zeros. The continuous critical-line cancellation is therefore literally destructive interference of phasors — the picture of Paper 8, now a confirmed computation. (*The numerical cancellation at the classical zeros: Fact. “Mechanism is phasor interference”: Reading, grounded.*)

The uniform (roots-of-unity) cancellation does **not** deform into the zeta cancellation. They are the same *mechanism* — interference nulls — one trivial and uniform, one hard and logarithmic. No derivation of one from the other is claimed.

3. The connective tissue (what the bridge genuinely gives)

1. **The spin rates are the forgetting-costs.** The phases are $t \cdot \ln n$, and $\ln n$ is the resistance — the Landauer erasure cost cataloged all cycle (the merge-erasure $\ln(n+1)$, etc.).

The integers in the zeta cancellation spin at rates equal to their erasure costs; the zeros are interference nulls of integers turning at their forgetting-costs. *(That $\ln n$ is both: Fact. The identification: the content.)*

2. **The genesis bit sits at the convergence boundary.** The bouquet converges on the critical line only because of the alternating sign $(-1)^{n-1} = e^{i\pi(n-1)}$ — the nontrivial character mod 2, the 2-phase — with the eta/zeta factor literally $(1 - 2^{1-s})$. The genesis bit, source of the first discrete cancellation $1 + (-1) = 0$, is also what tames the divergent integer bouquet into a convergent one: it appears at both ends, the seed of cancellation and the boundary of convergence. *(Alternation = 2-phase: Fact. “Both ends” framing: Reading.)*

4. Grading and the firm boundary

- **Fact** — roots of unity sum to zero; the log bouquet cancels at the classical zeros and not between; the alternation is the character mod 2 (the 2-phase).
- **Reading** — the cancellation mechanism is one and the same across discrete and continuous; the spin rates are the catalog’s forgetting-costs; the genesis bit bookends the picture (cancellation seed and convergence boundary).
- **Lead (deep, flagged)** — whether other prime “characters” (Dirichlet L-functions, the corpus’s wave/character layer, Paper 4) play roles analogous to the 2-phase. **No leverage on the zeros is claimed or implied**, and this should not be pushed to pretend otherwise.

Classical content: roots of unity, the Dirichlet eta function and its relation to zeta, the location of the nontrivial zeros, and conditional convergence on the critical line — all standard. The contribution is only the identification of the spin rates with the erasure costs and of the convergence-enabling alternation with the genesis-bit phase, and the confirmation that the interference picture is literal. No open arithmetic question is touched.

Originating direction (the Weyl-clock-cancellation bridge as the disciplined first step toward the critical line, kept strictly within the no-leverage boundary) is Alexander Pisani’s; the experiment, the connective-tissue observations, and this compilation were developed jointly with Claude (Anthropic). June 2026.